

torch.acosh#

<https://pytorch.org/docs/stable/generated/torch.acosh.html#torch.acosh>

Description:

Returns a new tensor with the inverse hyperbolic cosine of the elements of input. The domain of the inverse hyperbolic cosine is $[1, \infty)$ and values outside this range will be mapped to NaN, except for $+\infty$ for which the output is mapped to $+\infty$. input (Tensor) – the input tensor. out (Tensor, optional) – the output tensor.

Function Signature

```
torch.acosh(input: Tensor, *, out: Optional[Tensor]) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(4).uniform_(1, 2)
>>> a
tensor([ 1.3192, 1.9915, 1.9674, 1.7151 ])
>>> torch.acosh(a)
tensor([ 0.7791, 1.3120, 1.2979, 1.1341 ])
```

Notes & Warnings

NOTE

The domain of the inverse hyperbolic cosine is $[1, \infty)$ and values outside this range will be mapped to NaN, except for $+\infty$ for which the output is mapped to $+\infty$.

Detailed Documentation

Documentation

Returns a new tensor with the inverse hyperbolic cosine of the elements of input.

The domain of the inverse hyperbolic cosine is $[1, \text{inf})$ and values outside this range will be mapped to NaN, except for $+\text{INF}$ for which the output is mapped to $+\text{INF}$.

input (Tensor) – the input tensor.

out (Tensor, optional) – the output tensor.

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